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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Load the fully constrained architectural space from Exercise 1
df = pd.read_csv('arch_space.csv')

# 2. Perform Stratified Random Sampling (n=100)
# Stratifying on D1 (Propulsion) and D3 (Motor Count) to ensure proportion
target_n = 100

# Calculate exact sample sizes per stratum based on population proportions
strata_counts = df.groupby(['d1', 'd3']).size()
strata_props = strata_counts / len(df)
sample_sizes = (strata_props * target_n).round().astype(int)

# Adjust for rounding errors to ensure exactly 100 samples
while sample_sizes.sum() < target_n:
    sample_sizes.iloc[sample_sizes.argmin()] += 1
while sample_sizes.sum() > target_n:
    sample_sizes.iloc[sample_sizes.argmax()] -= 1

# Extract the stratified sample
sample_df = pd.DataFrame()
for (d1_val, d3_val), size in sample_sizes.items():
    stratum = df[(df['d1'] == d1_val) & (df['d3'] == d3_val)]
    if size > 0:
        sample_df = pd.concat([sample_df, stratum.sample(n=size, random_state=42)])

sample_df = sample_df.sample(frac=1, random_state=42).reset_index(drop=True)

# 3. Format into vector representation strings for the docx
vectors = []
for _, row in sample_df.iterrows():
    d4_str = f"{int(row['d4_1'])}{int(row['d4_2'])}{int(row['d4_3'])}{int(row['d4_4'])}"
    d7_str = f"{int(row['d7_1'])}{int(row['d7_2'])}{int(row['d7_3'])}{int(row['d7_4'])}"
    vec = f"{int(row['d1'])} - {int(row['d2'])} - {int(row['d3'])} - {int(row['d4'])} - {int(row['d7'])}"
    vectors.append(vec)

print(f"--- Generated {len(vectors)} Stratified Random Architectures ---")
for i, v in enumerate(vectors):
    print(f"Arch {i+1:03d}: {v}")

# Save the 100 N random sample to CSV for submission
pd.DataFrame(vectors, columns=['Vector_Representation']).to_csv('stratified_random_architectures.csv')

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# 4. Generate the Quantitative Justification Plot (Comparing Frequencies)
# Prepare data for plotting
decisions_to_plot = ['d1', 'd3', 'd6']
plot_titles = ['Decision 1 (Topology)', 'Decision 3 (Motors)', 'Decision 6 (Sensors)']

fig, axes = plt.subplots(1, 3, figsize=(15, 5), sharey=True)
fig.suptitle('Comparison of Rates of Occurrence: Population vs. Random Sample')

for i, (col, title) in enumerate(zip(decisions_to_plot, plot_titles)):
    # Calculate percentages
    pop_pct = df[col].value_counts(normalize=True).sort_index() * 100
    samp_pct = sample_df[col].value_counts(normalize=True).sort_index()

    # Combine into a single dataframe for seaborn
    plot_df = pd.DataFrame({
        'Alternative': pop_pct.index.tolist() + samp_pct.index.tolist(),
        'Percent (%)': pop_pct.tolist() + samp_pct.tolist(),
        'Dataset': ['Population'] * len(pop_pct) + ['Random Sample'] * len(samp_pct)
    })

    sns.barplot(data=plot_df, x='Alternative', y='Percent (%)', hue='Dataset', ax=axes[i])
    axes[i].set_title(title)
    axes[i].set_ylim(0, 100)

    # Add percentage labels on top of bars
    for container in axes[i].containers:
        axes[i].bar_label(container, fmt='%.1f%%', padding=3)

plt.tight_layout()
plt.show()

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--- Generated 100 Stratified Random Architectures ---
Arch 001: 2 - 2 - 0 - 010000 - 0 - 1 - 010201 - 1
Arch 002: 1 - 2 - 1 - 100001 - 2 - 1 - 010220 - 0
Arch 003: 1 - 0 - 2 - 110101 - 2 - 0 - 012011 - 1
Arch 004: 1 - 0 - 0 - 000001 - 1 - 0 - 010211 - 1
Arch 005: 1 - 1 - 0 - 001000 - 2 - 1 - 011221 - 0
Arch 006: 1 - 1 - 0 - 010000 - 0 - 0 - 012120 - 1
Arch 007: 0 - 0 - 1 - 010100 - 1 - 0 - 010221 - 0
Arch 008: 2 - 0 - 0 - 010000 - 0 - 1 - 011112 - 1
Arch 009: 0 - 2 - 1 - 000101 - 0 - 0 - 001202 - 0
Arch 010: 0 - 1 - 0 - 010000 - 0 - 0 - 012120 - 1
Arch 011: 0 - 1 - 1 - 100100 - 0 - 1 - 012002 - 1
Arch 012: 0 - 0 - 2 - 101101 - 2 - 1 - 012221 - 1
Arch 013: 1 - 1 - 2 - 110101 - 2 - 1 - 011221 - 0
Arch 014: 0 - 1 - 2 - 110101 - 0 - 1 - 001001 - 0
Arch 015: 2 - 0 - 1 - 101000 - 1 - 1 - 010120 - 1
Arch 016: 0 - 1 - 0 - 001000 - 0 - 1 - 001001 - 0
Arch 017: 1 - 1 - 2 - 011101 - 0 - 1 - 011022 - 1
Arch 018: 1 - 1 - 1 - 101101 - 0 - 1 - 010100 - 0
Arch 019: 1 - 1 - 1 - 101101 - 0 - 1 - 010100 - 0
Arch 020: 1 - 1 - 1 - 101101 - 0 - 1 - 010100 - 0

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Arch 018: 1 - 1 - 2 - 101101 - 2 - 1 - 012102 - 0
Arch 019: 0 - 1 - 1 - 110000 - 0 - 0 - 001221 - 0
Arch 020: 0 - 0 - 2 - 110101 - 2 - 0 - 012011 - 1
Arch 021: 1 - 0 - 1 - 010100 - 0 - 0 - 011122 - 1
Arch 022: 2 - 2 - 1 - 011000 - 2 - 1 - 010122 - 1
Arch 023: 0 - 1 - 1 - 010100 - 0 - 0 - 010221 - 1
Arch 024: 1 - 2 - 0 - 000001 - 2 - 1 - 010020 - 1
Arch 025: 1 - 0 - 2 - 101101 - 2 - 1 - 012221 - 1
Arch 026: 0 - 0 - 1 - 010001 - 2 - 0 - 011202 - 1
Arch 027: 1 - 0 - 0 - 000100 - 2 - 1 - 012221 - 1
Arch 028: 2 - 0 - 1 - 011000 - 2 - 0 - 000122 - 1
Arch 029: 0 - 1 - 0 - 000100 - 2 - 1 - 012102 - 0
Arch 030: 1 - 1 - 2 - 110101 - 0 - 1 - 001001 - 0
Arch 031: 0 - 0 - 1 - 001001 - 0 - 0 - 011001 - 1
Arch 032: 1 - 1 - 0 - 000001 - 0 - 1 - 011022 - 1
Arch 033: 2 - 1 - 0 - 001000 - 1 - 1 - 010212 - 1
Arch 034: 0 - 2 - 1 - 001100 - 0 - 1 - 001112 - 0
Arch 035: 2 - 2 - 1 - 001100 - 1 - 1 - 011112 - 0
Arch 036: 0 - 1 - 0 - 001000 - 2 - 1 - 011221 - 0
Arch 037: 1 - 2 - 1 - 100100 - 2 - 0 - 012021 - 1
Arch 038: 1 - 1 - 1 - 010100 - 0 - 0 - 010221 - 1
Arch 039: 0 - 0 - 2 - 011101 - 1 - 0 - 010211 - 1
Arch 040: 0 - 0 - 1 - 010100 - 0 - 0 - 011122 - 1
Arch 041: 1 - 2 - 1 - 000101 - 0 - 0 - 001202 - 0
Arch 042: 0 - 1 - 2 - 110101 - 2 - 1 - 011221 - 0
Arch 043: 0 - 2 - 0 - 000001 - 1 - 0 - 012021 - 0
Arch 044: 2 - 0 - 1 - 100100 - 1 - 0 - 012212 - 1
Arch 045: 0 - 2 - 1 - 100100 - 2 - 0 - 012021 - 1
Arch 046: 0 - 1 - 1 - 101000 - 2 - 0 - 012102 - 1
Arch 047: 2 - 2 - 0 - 000100 - 2 - 0 - 000120 - 1
Arch 048: 0 - 0 - 1 - 100001 - 1 - 0 - 010200 - 1
Arch 049: 1 - 2 - 1 - 001001 - 2 - 1 - 012100 - 1
Arch 050: 0 - 2 - 1 - 011000 - 2 - 1 - 012222 - 0
Arch 051: 0 - 1 - 1 - 100100 - 0 - 0 - 011121 - 0
Arch 052: 0 - 2 - 0 - 000001 - 2 - 1 - 010020 - 1
Arch 053: 0 - 1 - 1 - 000101 - 2 - 1 - 001102 - 0
Arch 054: 0 - 1 - 2 - 101101 - 2 - 1 - 012102 - 0
Arch 055: 0 - 1 - 0 - 000001 - 0 - 1 - 011022 - 1
Arch 056: 2 - 1 - 0 - 010000 - 1 - 0 - 012000 - 0
Arch 057: 0 - 0 - 0 - 000001 - 1 - 0 - 010211 - 1
Arch 058: 1 - 0 - 1 - 100001 - 1 - 0 - 010200 - 1
Arch 059: 0 - 2 - 2 - 011101 - 1 - 0 - 012021 - 0
Arch 060: 2 - 1 - 1 - 101000 - 2 - 1 - 001012 - 0
Arch 061: 1 - 1 - 1 - 000101 - 2 - 1 - 001102 - 0
Arch 062: 2 - 2 - 2 - 111100 - 0 - 0 - 012122 - 0
Arch 063: 1 - 0 - 1 - 010001 - 2 - 0 - 011202 - 1
Arch 064: 1 - 1 - 0 - 001000 - 0 - 1 - 001001 - 0
Arch 065: 1 - 0 - 1 - 001001 - 0 - 0 - 011001 - 1
Arch 066: 1 - 2 - 1 - 001100 - 0 - 1 - 001112 - 0
Arch 067: 1 - 2 - 0 - 000001 - 1 - 0 - 012021 - 0
Arch 068: 1 - 1 - 2 - 111001 - 0 - 0 - 012120 - 1
Arch 069: 1 - 0 - 1 - 010100 - 1 - 0 - 010221 - 0
Arch 070: 2 - 1 - 1 - 010100 - 0 - 0 - 001021 - 0
Arch 071: 2 - 2 - 0 - 010000 - 0 - 1 - 011210 - 0

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Arch 072: 1 - 0 - 0 - 001000 - 2 - 0 - 012011 - 1
Arch 073: 1 - 1 - 1 - 101000 - 2 - 0 - 012102 - 1
Arch 074: 1 - 1 - 0 - 000100 - 2 - 1 - 012102 - 0
Arch 075: 2 - 0 - 2 - 111100 - 1 - 1 - 012220 - 1
Arch 076: 1 - 1 - 1 - 100100 - 0 - 1 - 012002 - 1
Arch 077: 1 - 2 - 2 - 011101 - 1 - 0 - 012021 - 0
Arch 078: 0 - 2 - 2 - 011101 - 2 - 1 - 010020 - 1
Arch 079: 2 - 0 - 1 - 010100 - 1 - 0 - 010212 - 0
Arch 080: 1 - 0 - 1 - 100100 - 2 - 1 - 011222 - 1
Arch 081: 1 - 1 - 1 - 100100 - 0 - 0 - 011121 - 0
Arch 082: 2 - 1 - 0 - 010000 - 1 - 1 - 010200 - 1
Arch 083: 0 - 1 - 2 - 011101 - 0 - 1 - 011022 - 1
Arch 084: 0 - 1 - 2 - 111001 - 0 - 0 - 012120 - 1
Arch 085: 0 - 0 - 0 - 000100 - 2 - 1 - 012221 - 1
Arch 086: 1 - 2 - 1 - 011000 - 2 - 1 - 012222 - 0
Arch 087: 0 - 1 - 1 - 101000 - 1 - 0 - 012011 - 1
Arch 088: 0 - 0 - 0 - 001000 - 2 - 0 - 012011 - 1
Arch 089: 0 - 2 - 1 - 001001 - 2 - 1 - 012100 - 1
Arch 090: 2 - 2 - 1 - 011000 - 2 - 0 - 011022 - 1
Arch 091: 2 - 0 - 1 - 011000 - 0 - 1 - 010011 - 1
Arch 092: 1 - 0 - 2 - 011101 - 1 - 0 - 010211 - 1
Arch 093: 2 - 1 - 1 - 110000 - 1 - 1 - 012200 - 0
Arch 094: 2 - 1 - 0 - 001000 - 1 - 1 - 001002 - 1
Arch 095: 0 - 0 - 1 - 100100 - 2 - 1 - 011222 - 1
Arch 096: 1 - 1 - 1 - 101000 - 1 - 0 - 012011 - 1
Arch 097: 1 - 2 - 2 - 011101 - 2 - 1 - 010020 - 1
Arch 098: 0 - 2 - 1 - 100001 - 2 - 1 - 010220 - 0
Arch 099: 2 - 0 - 1 - 101000 - 0 - 1 - 012002 - 0
Arch 100: 1 - 1 - 1 - 110000 - 0 - 0 - 001221 - 0

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Comparison of Rates of Occurrence: Population vs. Random Sample



